Enhancing Intrusion Detection via Interpretable Inter-Flow Spatio-Temporal Graphs and Intra-Flow Features

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***Abstract*—The performance of network intrusion detection systems (IDS) critically depends on the completeness of traffic feature representation. Existing methods predominantly focus either on intra-flow features (e.g., packet length sequences) or single-dimensional inter-flow relationships, limiting optimal classification. To address this restriction, we propose IF^2-STG to enhance intrusion detection via interpretable inter-flow spatio-temporal graphs and intra-flow features. The key innovation of our scheme is the construction of an Inter-flow Spatio-Temporal relationship Graph (ISTG), with two-dimensional edges explicitly quantifying the correlation strength between flows. On this basis, a GNN learns the inter-flow context. In parallel, a CNN extracts fine-grained intra-flow features from raw bytes of individual flows. Finally, the two modal features are fused to acquire a comprehensive representation. This representation can effectively capture both intrinsic behavior patterns of flows and contextual inter-flow relationships, thereby significantly enhancing the detection performance and robustness of model. Experimental results show that the proposed IF^2-STG achieves over 99% accuracy on three real-world datasets, outperforming existing approaches. The constructed ISTG provides explicit visual interpretation of flow dependencies. More importantly, IF^2-STG-Inter, using only inter-flow features, maintains 97.78% detection accuracy in cross-dataset tests involving time-disparate datasets, validating superior generalization of inter-flow spatio-temporal features. This advantage renders it particularly promising for practical intrusion detection applications.**

***Index Terms*—Intra-flow features, inter-flow features, inter-flow spatio-temporal relationship graph, comprehensive representation, intrusion detection systems.**

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I. Introduction

NETWORK Intrusion Detection Systems (IDS) are a critical line of defense in cybersecurity, and their performance is vital for safeguarding network systems against attacks [1], [2], [3]. The efficacy of an IDS hinges on the completeness of its feature representation [4], [5], [6]. Network traffic inherently comprises multi-modal data with diverse features, which are broadly categorized into intra-flow and inter-flow features [54]. Intra-flow features describe the internal attributes of individual network flows, including statistical features (e.g., mean packet length, flow duration) [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], metadata features [23], [24], [25], [26], and sequence features (e.g., packet length sequences) [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37]. In contrast, inter-flow features mainly refer to the relationship features among network flows.

In existing IDSs, most systems focus solely on a single type of intra-flow feature [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], making them highly sensitive to minor variations in network flow features. For instance, when facing attacks where adversaries tamper with packet lengths [10], [11], IDSs based on sequence features (e.g., packet length sequences) often struggle to maintain stable performance, exhibiting poor robustness. Consequently, some studies have adopted multi-modal learning approaches using diverse network flow features [41], [42], [43], [44], [45], [46], [47], [48], [49]. However, their feature selection remains confined to intra-flow features, neglecting the inter-flow correlations [54], [63].

To address the limitations of intra-flow features, researchers have attempted to utilize multiple network flows to learn inter-flow relationships for malicious traffic detection [41], [50], [51]. Among them, CBSeq [50] treats all flows between two hosts as an integrated entity, concatenating features like packet count, time intervals, and source ports of each flow to construct behavioral sequences for classification. Similarly, HDFEF [51] maps features of individual flows into different spaces and combines these multi-space features to form composite multi-flow representations. However, these methods only integrate information from multiple flows through feature aggregation, failing to deeply explore the underlying temporal dependencies and spatial correlations among flows.

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In fact, network attacks often exhibit distinct spatio-temporal correlation patterns. For example, in a Distributed Denial of Service (DDoS) attack, traffic originating from multiple source IPs coordinates high-frequency requests temporally while forming distributed assaults on target servers spatially [12]. To capture these characteristics, GAT_based model [52] and MeDF [54] construct graphs from temporal and spatial perspectives, respectively. But they can only model inter-flow dependencies in a single dimension. Although recent studies such as PPT-GNN [65] and CCSTGN [66] employ spatio-temporal graph networks to capture the spatio-temporal dynamics among flows, they cannot explicitly quantify the strength of inter-flow correlations, limiting interpretability.

Furthermore, most existing studies focus solely on either intra-flow or inter-flow features. Although some recent approaches have attempted to jointly learn both types of features, they still have significant limitations in mining inter-flow relationships. For instance, MeDF [54] only constructs graphs from the spatial dimension, ignoring temporal dependencies between flows. IIT [63] implicitly learns inter-flow dependencies through attention mechanisms, but lacks an explicit graph structure, failing to visually reveal the spatio-temporal correlations.

In light of this, this paper proposes a novel representation learning paradigm for intrusion detection (IF^2-STG),[1] designed to enhance model robustness by jointly modeling fine-grained intra-flow content and explicit inter-flow spatio-temporal correlations. Its core innovation lies in an interpretable Inter-flow Spatio-Temporal relationship Graph (ISTG), which quantifies dependencies between flows using edges defined by start time intervals and IP-set associations. IF^2-STG employs two parallel learning branches: a CNN for capturing intra-flow features from raw bytes, and a GNN for modeling inter-flow relationships via the ISTG. Their complementary representations are then fused into a comprehensive flow representation. Compared to existing methods, this work is the first to explicitly quantify and structurally model complex spatio-temporal dependencies among flows, enabling interpretable analysis of multi-flow attack patterns while demonstrating stronger generalization and robustness against label noise.

The main contributions of this work are summarized as follows:

- We propose a novel intrusion detection model that integrates inter-flow spatio-temporal relationships with intra-flow features. The model employs CNN and GNN architectures to learn intra-flow and inter-flow feature representations respectively, then fuses them into a comprehensive flow representation. This unified representation preserves intrinsic behavioral patterns of individual flows while incorporating contextual spatio-temporal dependencies among multiple flows, thereby significantly improving classification accuracy and robustness.
- We construct an ISTG to capture the correlations among flows in both temporal and spatial dimensions. Using packet length sequences and statistical values as node features and defining two-dimensional edge vectors based on the time intervals and IP sets associations, ISTG can explicitly model the spatio-temporal dependencies between flows. The proposed ISTG demonstrates good interpretability and representational capability, particularly for attack types with significant inter-flow interaction patterns such as DDoS and port scanning.
- Our proposed IF^2-STG achieves over 99% accuracy on three real-world datasets, validating the effectiveness of the joint learning of intra-flow and inter-flow features. Cross-dataset experiments further reveal that inter-flow spatio-temporal features exhibit remarkable generalization across datasets collected at different times, offering a valuable solution against the evolving attacks in the real-world.

The rest of this paper is organized as follows. Section II summarizes related work and analyzes existed methods. Section III explains the research motivation. Section IV introduces the detailed design of the proposed IF^2-STG. Section V presents the experimental results and performance evaluation. Finally, Section VI concludes the paper.

II. Related Work

The primary objective of an IDS is to accurately classify network traffic and discriminate malicious behavior [7], [8], [9]. Feature extraction is a crucial step in this process. Existing research can be primarily categorized into two classes based on the scope of feature extraction: intra-flow feature-based methods and inter-flow feature-based methods. To clarify the distinctions between existing methods and our approach, Table I provides a detailed comparison in terms of the features used, features extracted, models, and limitations.

A. Intra-Flow Feature-Based Methods

IDSs based on intra-flow features identify threats by learning the inherent characteristics within individual network flows. These methods can be categorized into single-modal learning and multi-modal learning approaches.

1) Single-Modal Learning: Single-modal learning methods utilize only one type of intra-flow feature, including statistical features, metadata features, and sequence features.

Statistical feature-based methods rely on statistical features extracted from bidirectional flows (e.g., total packet count, packet inter-arrival time, flow duration, etc.) [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [80], combined with machine learning algorithms to achieve traffic classification and recognition [15]. Yang et al. [13] introduced Principal Component Analysis to account for the correlations between different statistical attributes, and used sliding window to achieve scalable anomaly detection. Apruzzese et al. [17] utilized features such as total transmitted packets, total transmitted bytes, and bytes per second to propose a Random Forest (RF)-based IDS with certain defenses against adversarial attacks. Bozkır et al. [18] employed the NFStream tool to compute traffic features and designed an encrypted traffic classification platform using GB-Tree, LightGBM, and XGBoost algorithms. Similarly, Wang et al. [19] extracted statistical features, including flow duration,

[1] The source code is publicly available at https://github.com/Akap-vv/IF2-STG.

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TABLE I
OVERVIEW OF PRIOR RESEARCH AND THIS WORK

Method	Features used	Features extracted				Model	Limitations
		Intra-flow	Inter-flow				
			Aggregation	Temporal	Spatial		
[13]	STA	✓				LSHiForest	Inadequate feature representation; Sensitive to feature perturbations
[17]		✓				RF	
[18]		✓				GBTree	
[19]		✓				XGBoost	
[23]	Metadata feature	✓				MLP,CNN,LSTM	
[24]		✓				DT,SVM,RF	
[25]		✓				CNN	
[27]	PLS	✓				LSTM	
[28]		✓				GRU	
[30]		✓				GNN	
[31], [32], [33]	RBS	✓				CNN	
[34]		✓				Transformer	
[35]		✓				CNN-LSTM	
[36]		✓				Transformer	
[37]		✓				GNN	
[42]	STA	✓				MDAE	Limited to intra-flow analysis; Neglect inter-flow correlations
[43]	STA	✓				CNN-GRU	
[44]	PLS,PWSS,PFS	✓				Multi-kernel convolution	
[48]	PBS,PLS	✓				CNN-LSTM	
[49]	STA	✓				CNN-LSTM	
[41]	RBS,PLS,IATS		✓			EdgeNet	Simple feature aggregation
[50]	STA,PLS,IATS		✓			Transformer	
[51]	RBS		✓			CNN-LSTM	
[52]	STA			✓		GAT	Lack spatio-temporal modeling
[53]	STA			✓		GCN	
[54]	RBS	✓			✓	GCN	
[55]	STA				✓	ProGraph	
[58]	STA			✓		GCN	Lack explicit graph structure
[60]	STA			✓		Flowformer	
[61]	RBS			✓		LSTM	
[64]	PLS,IATS			✓		Transformer	
[66]	STA			✓	✓	GNN	High computational overhead Lack interpretability
[67]	STA			✓	✓	TGN	
Our	RBS,PLS	✓		✓	✓	CNN-GNN	–

RBS: raw byte sequences; PLS: packet length sequences; IATS: inter-arrival time sequences; STA: statistical features; PWSS: packet window size sequences; PFS: packet TCP flag sequences; RF: random forest; MLP: multi-layer perceptron; CNN: convolutional neural network; LSTM: long short-term memory; DT: decision tree; SVM: support vector machine; GRU: gated recurrent unit; GNN: graph neural network; GCN: graph convolutional network; TGN: temporal graph network; MDAE: multimode deep auto-encoder;

time intervals, and bytes per second, from encrypted traffic and applied an XGBoost model for classification. Ngo et al. [80] conducted a comparative study of machine learning methods for threat detection in healthcare IoT systems based on statistical features.

Metadata feature-based methods directly extract raw fields from packet headers without requiring statistical information [23], [24], [25], [26], which can accelerate detection speed and enhance real-time performance. Ma et al. [23] and Li et al. [24] both extract frame and protocol metadata features from packets, and incorporate feature selection methods to identify the most informative features, thereby reducing the computational overhead. Likewise, the Lucid [26] approach extracts 11 key metadata features from each packet and utilizes a lightweight CNN to achieve real-time DDoS detection.

Sequence feature-based methods focus on temporal patterns in network traffic, leveraging deep learning (DL) models to capture the trend and periodic characteristics of sequences [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37]. The input data for these models typically encompass packet length sequences, inter-arrival time sequences, and raw byte sequences. ERNN [27], specifically designed for network-induced phenomena, utilizes LSTM networks to classify packet length sequences. FS-Net [28] employs a bidirectional GRU to encode packet length sequences, incorporating a reconstruction mechanism to ensure feature representation validity. Han et al. [30] constructed graphs based on packet length sequences using GNN to enhance detection capabilities. Raw bytes-based methods learn from the hexadecimal byte sequences of flows, typically converting byte sequences into grayscale images or textual representations, and then extracting features through models like CNN and Transformer. Studies [31], [32], [33] all transformed hexadecimal bytes into grayscale images and applied CNN models for classification. Meng et al. [34] employed a Transformer to encode byte sequences and then use natural language processing (NLP) techniques for classification. Gao et al. [35] treated discrete hexadecimal bytes as words and integrated attention mechanisms into CNN-BiLSTM to enhance IDS's accuracy by focusing on key information. Zhao et al. [36] adopted a multi-level flow representation and the Traffic Transformer model to better extract features from raw byte data. Luo et al. [37] constructed a flow-level semantic graph from raw byte

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data and aggregated it into semantic embeddings using a gated GNN.

The aforementioned approaches suffer from two major limitations. First, the features extracted solely from individual flows may be incomplete, failing to provide sufficient information for accurate attack prediction, especially for multi-stage attacks [38], [39], [40]. Second, these methods are sensitive to minor perturbations in single-flow features, making them vulnerable to evasion attacks. For instance, as demonstrated in [10], an evasion strategy that modifies packet lengths in malicious traffic to mimic benign traffic characteristics can significantly degrade the detection accuracy of IDS relying solely on packet length sequences.

2) Multi-Modal Learning: Multimodal learning can enhance performance by capturing patterns from multiple perspectives, and it can alleviate the problems of insufficient representation and vulnerability to attacks that arise from relying solely on single features [41], [42], [43], [44], [45], [46], [47], [48], [49]. Kong et al. [41] employed a sliding sampling window to conduct real-time sampling of network conversations and extracted three sequential features—bytes, lengths, and time intervals—to capture the patterns of different types of traffic. He et al. [42] proposed a novel multi-view solution to reduce feature complexity and employed multimodal DL to construct effective feature fusion modules for learning the underlying structure of traffic data. Yu et al. [43] designed a DL method based on multi-view learning, utilizing an ensemble classifier for multi-class classification of minority attacks. Yun et al. [44] extracted three attribute sequences from TLS flows and used DL models to capture long-term dependencies among elements in packet sequences. Wang et al. [48] proposed a multimodal encrypted traffic classification framework, AppNet, which uses a 1D CNN to extract features from the first 1014 bytes of the initial packet and an LSTM to learn temporal relationships in packet length sequences, ultimately concatenating features learned from both perspectives for classification. Lei et al. [49] proposed a multi-feature correlation analysis method to capture relationships between different feature types, and utilized a CNN-LSTM model for spatiotemporal feature extraction.

Nevertheless, the above methods remain limited to analyzing the internal features of individual flows while failing to account for inter-flow correlations, which is not conducive to detecting sophisticated attack patterns involving interactions among multiple flows.

B. Inter-Flow Feature-Based Methods

IDSs leveraging intra-flow features discover malicious activities by analyzing correlations among multiple network flows, which can be categorized into two types: flow-level feature aggregation and inter-flow relationship mining.

1) Flow-Level Feature Aggregation: The essence of flow-level feature aggregation methods is the cross-flow concatenation of single-flow features. For example, both iDetector [41] and CBSeq [50] take multiple flows between two hosts as a whole, where the former extracts the byte sequences, packet length sequences, and inter-arrival time sequences of all packets for classification, and the latter constructs behavior sequences by combining the packet counts, intervals, and source/destination port of multiple flows. HDFEF [51] projects individual flow features into different latent spaces and forms combined features by concatenating multi-space features. However, these methods simply aggregate the raw or transformed features of multiple flows, without considering the temporal order or spatial topological correlations between flows.

2) Inter-Flow Relationship Mining: Inter-flow relationship mining methods learn correlation features between flows through either explicit graph construction or implicit dependency modeling [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [79]. Ngo et al. [79] constructed graphs based on Top-K attribute similarity and adopted the GraphSAGE model to effectively capture node representations for IoT intrusion detection. Xu et al. [52] and Peng et al. [53] segmented raw traffic into blocks, treated flows as nodes, and established edges based on temporal cross-correlations. They employed graph attention networks (GAT) and graph convolutional networks (GCN), respectively, to extract node representations by aggregating information from adjacent nodes. Wang et al. [54] proposed a graph-based approach for inter-flow feature extraction, where individual flows were represented as nodes and edges were established based on IP set correlations and protocol consistency, subsequently employing GCN to learn spatial relationships among flows. Li et al. [55] proposed ProGraph, a graph-propagation-based method. It builds a correlation graph with session clusters aggregated from different networks, upon which effective graph propagation can be implemented iteratively to predict labels for test nodes. These methods lack joint modeling of both temporal and spatial dimensions in inter-flow relationships.

Zhao et al. [58] built both attribute and temporal relation graphs for flow sequences. But in temporal relationship extraction, the edge weights are determined solely by the ordinal position differences between flows, failing to accurately capture the true time-dependent intensity. MT-FlowFormer [59] extracts deep-level features from the flow sequence, which can effectively distinguish the relevance and importance of different flows in the flow sequence. Yao et al. [60] and Vaswani et al. [61] treated the stream sequence as a time sequence and used attention-based LSTM to learn the temporal relationships among network flows. Meng et al. [63] represented flows using packet length sequences and inter-arrival time sequences, employing attention mechanisms to implicitly learn inter-flow dependencies. Nevertheless, the absence of an explicit graph structure in their approach hindered intuitive visualization of flow relationships.

The recently proposed PPT-GNN [65] and CCSTGN [66] methods leverage graph structures to capture spatio-temporal dynamics of cyber attacks, achieving significant advances in real-time performance and generalization. However, PPT-GNN constructs all flows within a time window into a single graph, which may incur significant computational overhead. Moreover, they fall short in showcasing explicit inter-flow spatio-temporal patterns within the graph, limiting interpretability.

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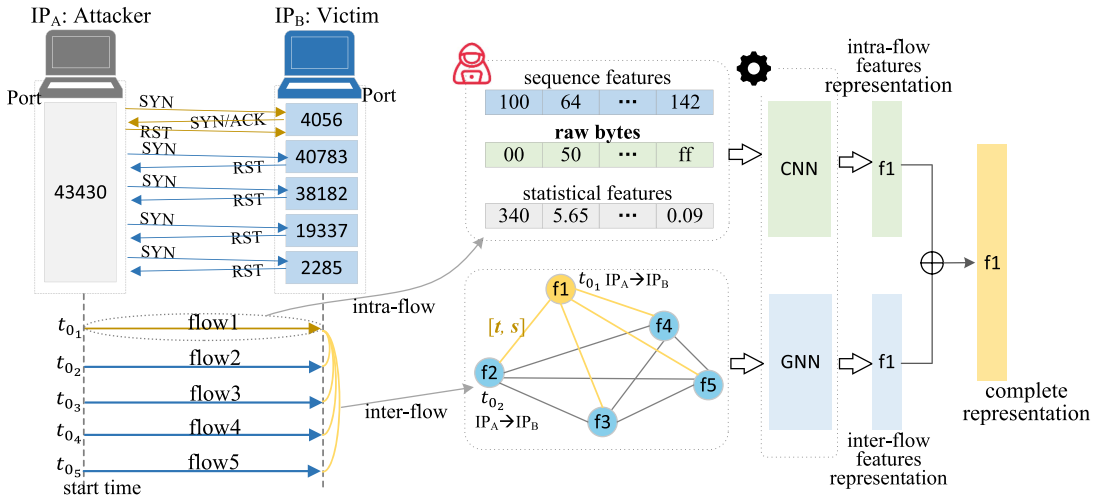


Fig. 1. Research motivation. Taking scan attacks as an example: it is difficult to accurately identify the attack intent relying solely on the internal features of a single flow. Thus, inter-flow spatio-temporal relationships are combined with intra-flow features to form a comprehensive representation.

It is worth noting that frameworks integrating intra- and inter-features have also been explored in broader cybersecurity contexts, such as insider threat detection [78]. However, this framework processes user behavior logs and models relationships based on organizational structures. Since both the data objects and relational semantics differ fundamentally from those in network intrusion detection, it cannot be directly adapted for modeling spatio-temporal correlations in network traffic.

To overcome the aforementioned limitations, this paper innovatively proposes the ISTG, which is designed to achieve effective spatio-temporal joint representation for the field of network intrusion detection, while providing interpretable insights. Furthermore, to compensate for the potential loss of fine-grained details in inter-flow analysis, our method incorporates intra-flow raw byte features for multi-modal fusion. By leveraging complementary information, this design strives to improve the accuracy and robustness of detection.

III. Motivation

This section elucidates the research motivation and necessity by taking half-open port scanning attacks as an example. The attacker sends SYN packets to target ports, determining a port as open upon receiving SYN/ACK responses or closed upon RST packets. As illustrated in the top-left corner of Fig. 1, attacker A sequentially probes five ports on target host B. During this process, each packet sequence between host A and every probed port of host B constitutes an independent bidirectional flow. The simplified diagram in the bottom-left corner represents these flows as arrows arranged vertically in chronological order of start time.

For $flow1$, relying solely on its intra-flow features (such as packet length sequences, raw bytes, statistical features, etc.) to determine whether it belongs to malicious traffic or normal (request) traffic has obvious defects. On the one hand, a single type of feature can be easily manipulated by attackers to evade detection; on the other hand, single-flow behavior lacks contextual correlation, making it impossible to identify the continuous port probing behavior of port scanning attacks. To overcome these deficiencies, apart from intra-flow features, we use graph representation learning to capture inter-flow relationships. Specifically, each graph node represents an individual flow, while edges denote correlations between flows. Taking $flow1$ and $flow2$ as examples, their start times are close, and both are flows from host A to host B in spatial topology. Such temporal and spatial correlations can be quantitatively represented as t and s respectively, forming a two-dimensional vector that serves as the edge attribute. Eventually, a spatio-temporal relationship graph containing 5 flows is constructed.

Subsequently, the intra-flow features and inter-flow spatio-temporal relationship graphs are separately learned via different networks to generate the intra-flow feature representation and inter-flow feature representation of $f1$. These two representations are then fused to form the comprehensive representation of $f1$ that captures both its internal behavior patterns and contextual relationships with other flows. This integrated representation offers the model a more comprehensive discriminative basis, thereby improving the accuracy of intrusion detection.

IV. Description of IF^2-STG

The proposed IF^2-STG comprises four components: data preprocessing module, intra-flow feature extraction module, inter-flow feature extraction module, and fusion classification module. The data preprocessing module extracts the features required by the intra-flow and inter-flow extraction modules. The intra-flow and inter-flow extraction modules process their input features independently to generate flow-level feature representations for the fusion module and classify them. These representations are then concatenated and classified in the fusion classification module to achieve comprehensive detection. The overall framework of the method is shown in Fig. 2.

In terms of feature selection for network flows, the intra-flow feature extraction module adopts raw bytes to preserve

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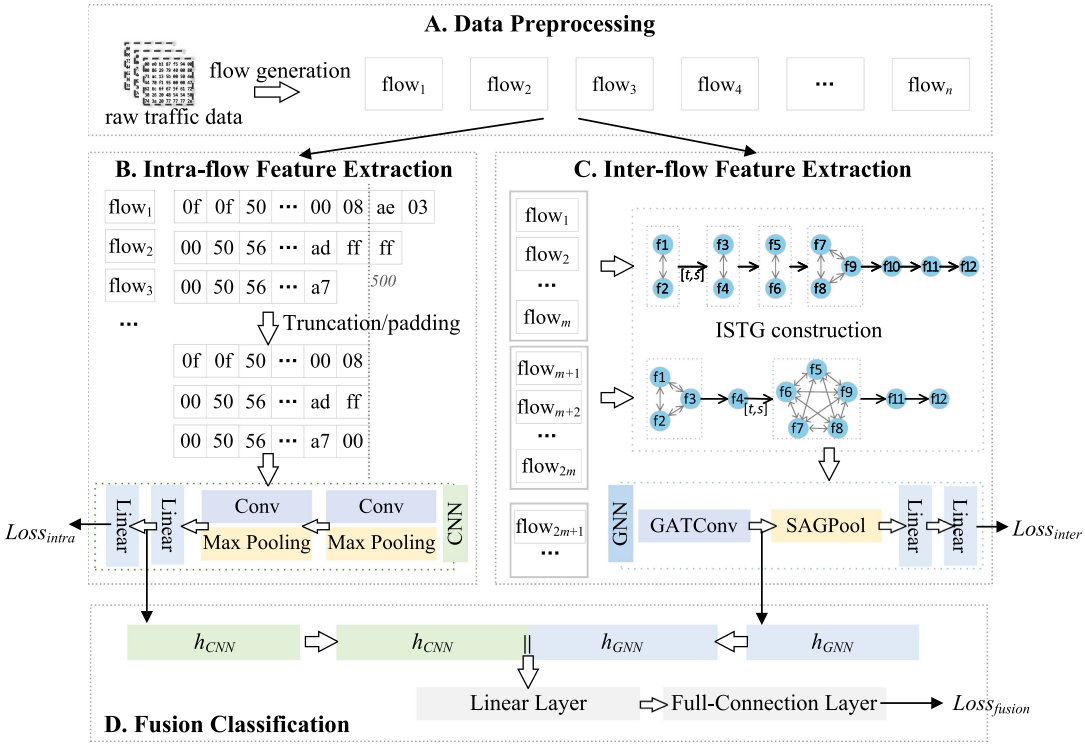


Fig. 2. The system framework of the proposed IF^2-STG, consisting of four modules: *(a) Data Preprocessing* provides features required for subsequent modules, *(b) Intra-flow Feature Extraction* and *(c) Inter-flow Feature Extraction* learn the representations of two modalities, and *(d) Fusion Classification* combines these representations for final intrusion detection.

fine-grained information. However, byte sequences typically need to be truncated into a fixed length for model input compatibility, inevitably resulting in information loss. In the inter-flow feature extraction module, the packet length sequences and their statistical features are used as node features to construct graphs. On the one hand, compared to using only basic triplets (source IP, destination IP, and protocol) as node features [52], the combination of packet length sequences and statistical features provides more comprehensive behavioral patterns of the flows themselves. On the other hand, this combination also makes up for the information loss caused by the truncation of raw bytes. Therefore, the selection of these multi-dimensional flow features helps to improve the integrity of traffic characterization and the robustness of the model.

A. Data Preprocessing Module

The raw traffic consists of a set of continuous and ordered data packets. First, packets are divided into corresponding bidirectional network flows based on their five-tuple (source IP, source port, destination IP, destination port, protocol). Each flow contains a series of temporally ordered data packets, denoted as $F_i = \{P_1, P_2, \ldots, P_n\}$, where P_j represents the j-th packet in the flow.

Subsequently, for each flow, the raw byte sequence and the packet length sequence are extracted for use in the subsequent intra-flow feature extraction module and inter-flow feature extraction module, respectively. The raw byte sequence of a flow is obtained by concatenating the raw bytes of each packet in order. For a packet P_j, its raw byte sequence is denoted as $byte_j = [b_{j1}, b_{j2}, \ldots, b_{jl_j}]$, where b_{jl_j} is the last byte, and $l_j = \text{len}(P_j)$ is the length of packet P_j. Therefore, the raw byte sequence of flow F_i is $B_i = byte_1||byte_2||\cdots||byte_n = [b_{11}, b_{12}, \ldots, b_{1l_1}, b_{21}, b_{22}, \ldots, b_{2l_2}, \ldots, b_{n1}, b_{n2}, \ldots, b_{nl_n}]$. The packet length sequence of F_i is $L_i = [l_1, l_2, \ldots, l_n]$.

To adapt to the model input, the raw byte sequence and packet length sequence are truncated to fixed lengths q and p, respectively, with zero-padding applied to sequences shorter than the specified lengths. Additionally, the timestamp t_0 of the first packet in each flow is taken as the start time of the flow, which will be used to compute the temporal component of edges in the spatio-temporal relationship graph within the inter-flow feature extraction module.

B. Intra-Flow Feature Extraction Module

For intra-flow feature extraction, we perform end-to-end learning on raw byte data. Network traffic is essentially sequential byte data, where the hierarchical structure of bytes, packets, sessions, and entire flows closely resembles the character, word, sentence, and document structure in NLP. In recent years, the successful applications of CNN in NLP all used 1D-CNN, for example, sentiment analysis and text classification [67], [68]. Therefore, we adopt a 1D-CNN architecture similar to [31]. The inputs to this network comprise the first q bytes of each flow, with zero-padding for shorter sequences. The input of the last

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TABLE II
THE KEY PARAMETERS OF 1D-CNN MODEL

Operator	Input	Filter size	Filter number	Stride	Pad	Output
conv2d	q*1	25*1	32	1	same	q*32
max pooling	q*32	3*1	–	3	same	167*32
conv2d	167*32	25*1	64	1	same	167*64
max pooling	167*64	3*1	–	3	same	56*64
linear	56*64	–	–	–	–	F_{CNN}
linear	F_{CNN}	–	–	–	–	C

linear layer is taken as the flow representation $h_{CNN} \in \mathbb{R}^{F_{CNN}}$, which is fed into the fusion classification module. Table II details the key parameters of each layer in the 1D-CNN model, where F_{CNN} denotes the dimension of the CNN's output hidden vector and C represents the number of flow categories in the multi-classification task.

C. Inter-Flow Feature Extraction Module

Multiple consecutive network flows exhibit inherent spatio-temporal correlations. Consequently, effective analysis should account for both the intrinsic attributes of individual flows and their inter-flow relationships. To this end, we employ GNN to model and learn the spatio-temporal correlations among network flows.

1) Construction of ISTG: For each network flow category, we take m consecutive network flows with adjacent start times as a group to construct an ISTG. In the graph, each node corresponds to an individual flow, where all nodes share identical labels that consequently define the graph's label. We select the packet length sequence and relevant statistical features as node features.

Packet length sequence feature: The packet length sequence has been widely used in network flow classification tasks, particularly for encrypted traffic where available features are limited [27], [28], [29], [30]. Since network flows are bidirectional, we distinguish forward and backward packets by assigning positive values to forward packet sizes and negative values to backward packet sizes in the sequence. To standardize input dimensions for the model, we select the first p packets of each flow (padding with zeros for shorter flows), though this truncation may omit subsequent packets in long flows. To compensate for this limitation, we supplement with statistical features.

Statistical feature: Following [52], we compute 18 statistical features separately for the entire flow, forward flow, and backward flow, including: maximum, minimum, mean, median absolute deviation, standard deviation, variance, skewness, kurtosis, and percentiles (10% to 90%) of packet sizes, along with packet counts. This yields 54 statistical features per flow. To reduce dimensionality, we employ random forest for feature importance ranking and select the top z most important features as complementary to the sequence features.

Thus, the node features can be expressed as $h = [l_1, l_2, \ldots, l_p, s_1, s_2, \ldots, s_z]$, where l_i and s_i denote the packet size and statistical feature values, respectively.

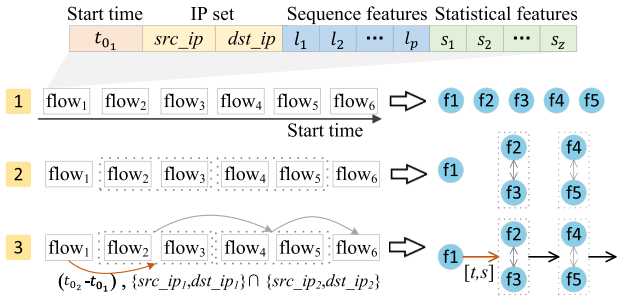


Fig. 3. An example of ISTG construction (assuming m=6), comprising three key steps: 1) Flow Sorting: Sorting the m flows by their start times, corresponding to the node order in ISTG. Each flow requires features such as start time, IP set, sequence features, and statistical features. 2) Concurrent Flow Grouping: Flows within each dashed box form a concurrent group, and bidirectional edges connect concurrent flows in ISTG. 3) Spatio-temporal Relationship Calculating: Nodes or groups in ISTG are sequentially linked with unidirectional edges weighted by 2D spatio-temporal vectors $[t, s]$.

Next is the critical design of the ISTG - the edge connections between nodes. We employ a two-dimensional feature vector as the edge attribute, with the two dimensions quantifying the temporal correlation and spatial correlation between flow pairs respectively. Specifically, suppose there is an edge $e_{ij} = [t, s]$ between node i and node j. The temporal correlation component t is derived from the start time difference between the two flows. Theoretically, flows with smaller start time intervals demonstrate stronger temporal correlations. We therefore implement this logic using an exponential decay function e^{-x}, which assigns higher t values to flow pairs with smaller time gaps. For the spatial correlation component s, we determine it by checking whether there is an intersection between the source IP and destination IP sets $IP = \{src_ip, dst_ip\}$ of both flows. The value s is set to 1 if any IP intersection exists, and 0 otherwise. This judgment method can effectively capture diverse network interaction patterns, such as multiple connections initiated from the same device, different devices accessing the same server, and chain-style communication.

Therefore, the weight of edge $e_{ij} = [t, s]$ can be formally computed as follows:

$$t = e^{-(t_{0_i} - t_{0_j})}, t_{0_i} \geq t_{0_j} \tag{1}$$

$$s = \begin{cases} 1, & \text{if } IP_i \cap IP_j \neq \emptyset \\ 0, & \text{else} \end{cases} \tag{2}$$

In addition, considering network bandwidth fluctuations, we regard flows with a time difference less than a specified threshold (10 ms) and an intersection in source/destination IPs as concurrent flows. These concurrent flows form a concurrent group, where any two nodes are connected by bidirectional edges with weights [1.0, 1.0]. Nodes between different groups, between a group and a single node, and between single nodes are all connected by unidirectional edges in chronological order. Additionally, in order to reduce the complexity of the ISTG, unidirectional edges that can be derived through transitivity are not explicitly generated. Fig. 3 illustrates an example of ISTG construction. Compared with the way of simply connecting

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sequentially in chronological order, this connection method considering concurrency demonstrates superior effectiveness, which will be verified in Section. V-F of our experiments.

2) Graph Representation Learning: The labeled ISTG is fed into a GNN for inter-flow feature extraction and graph classification. The GNN model we use primarily consists of a Graph Attention Layer (GAT) [69], a Self-Attention Graph Pooling Layer (SAGPool) [70], [71], and linear layers.

The GAT utilizes the self-attention mechanism to enable each node in the graph to dynamically aggregate information from its neighboring nodes based on their importance, thereby learning better node representations. For a given node i in the graph with feature vector h_i, its set of neighboring nodes is denoted as $\mathcal{N}(i)$. Taking neighboring node $j \in \mathcal{N}(i)$ as an example, GAT calculates the importance of node j to node i through the following formula:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(a^T[Wh_i \parallel Wh_j \parallel W_e e_{ij}]\right)\right)}{\sum_{k\in\mathcal{N}_i}\exp\left(\text{LeakyReLU}\left(a^T[Wh_i \parallel Wh_k \parallel W_e e_{ik}]\right)\right)} \tag{3}$$

where T represents transposition and $||$ is the concatenation operation. W is a weight matrix that is applied to every node, and W_e is a weight matrix that is applied to every edge. a is the parameter of a single-layer feedforward neural network. The LeakyReLU function [72] is introduced for non-linearization, and the Softmax function is employed to calculate the final α_{ij}.

Subsequently, the attention weights of neighboring nodes are linearly accumulated to update the feature vector $h_{i'}$ of node i.

$$h_{i'} = \sigma\left(\sum_{j\in\mathcal{N}_i}\alpha_{ij}Wh_j\right) \tag{4}$$

To stabilize the self-attention learning process, GAT employs a multi-head attention mechanism, where K independent attention mechanisms compute node features in parallel, and their outputs are concatenated:

$$h_{i'} = \|_{k=1}^{K}\,\sigma\left(\sum_{j\in\mathcal{N}_i}\alpha_{ij}^k W^k h_j\right) \tag{5}$$

where α_{ij}^k denotes the attention coefficient calculated by the k-th attention head, and W^k is the linear transformation matrix for the k-th attention head.

In this work, we use a three-head attention mechanism (K=3). The representations generated by each head are concatenated to obtain a new representation for each node (i.e., each flow), incorporating both its original information and contextual information from neighboring nodes. Consequently, we treat this as a flow-level representation embedding inter-flow relationships and output it to the subsequent fusion classification module, denoted as $h_{GNN} = h_{i'} \in \mathbb{R}^{F_{GNN}}$.

To capture the global structural information of the graph, this module performs graph classification rather than node classification, which only considers local neighborhood information of nodes. Specifically, to pay more attention to important nodes and improve training efficiency, we use a SAGPool mechanism to select important nodes and simplify the graph structure. Then, through a linear layer and a softmax function, the prediction probability of graph classification is obtained. Since the main purpose of this study is flow classification, the graph classification results are expanded by m times to obtain the node-level flow classification results.

D. Fusion Classification Module

This module takes the flow-level representations from both the intra-flow and inter-flow feature extraction modules and concatenates them to obtain the final representation vector $h_f = h_{CNN}||h_{GNN}$. Here, h_f integrates fine-grained intra-flow byte features and context-aware inter-flow relational features, providing a more comprehensive characterization of network flows. Subsequently, h_f is fed into a linear layer to learn and integrate the complex nonlinear relationships between the two types of features, with the final classification performed by a fully-connected layer.

E. Loss Function

In the fusion classification module, the motivation for concatenating h_{CNN} and h_{GNN} is to enable the neural network to combine the strengths of intra-flow and inter-flow features, allowing them to complement each other. However, due to the black-box nature of neural networks, it may not learn the best solution for both components as expected [73]. Therefore, we hypothesize that if the combined model's parameters are optimal, then the parameters of the two feature extraction modules should also be individually optimal. Based on this, we adopt multi-loss joint training to guide the model to achieve the best performance of each module. Specifically, the loss function of IF^2-STG incorporates classification losses from three modules, evaluating not only the final classification performance but also the classification capabilities of the features learned by the two feature extraction modules. To ensure balanced attention to both intra-flow and inter-flow features during model training, we assign equal weights to their respective loss functions.

$$Loss = Loss_{intra} + Loss_{inter} + Loss_{fusion} \tag{6}$$

$Loss_{intra}$, $Loss_{inter}$, and $Loss_{fusion}$ represent the classification losses of the intra-flow feature extraction module, inter-flow feature extraction module, and fusion classification module, respectively. The cross-entropy loss is applied to calculate the loss between the predicted value p and the ground truth y.

$$loss_{\text{intra}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_i \log\left(p_{ic}^{\text{intra}}\right) \tag{7}$$

$$loss_{\text{inter}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_i \log\left(p_{ic}^{\text{inter}}\right) \tag{8}$$

$$loss_{\text{fusion}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_i \log\left(p_{ic}^{\text{fusion}}\right) \tag{9}$$

Here, N denotes the total number of samples, and C represents the number of categories in the multi-class classification task.

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$p_{\bar{i}c}$ indicates the predicted probability from each module's classification layer that sample i belongs to class c. It should be noted that in the inter-flow feature extraction module, the GNN outputs graph-level classification results, whereas we focus on flow-level classification. Since all nodes within a graph should share the same class label, and considering that each graph contains m nodes, we expand each graph's prediction result by a factor of m to obtain node-level predictions.

Specifically, assuming the GNN outputs the predicted probability $p_k = [p_{k1}, p_{k2}, \ldots, p_{kC}] \in \mathbb{R}^C$ for graph G_k, where p_{kc} represents the probability of graph G_k belonging to class c. Replicate p_k for m times to generate the node-level probability matrix $P_k^{inter} \in \mathbb{R}^{m \times C}$. Finally, all node-level probability matrices are concatenated to form the complete flow-level prediction matrix $P^{inter} \in \mathbb{R}^{N \times C}$ for all N samples (flows), and $p_i^{inter} \in P^{inter}$.

V. Evaluation

A. Experiment Preparation

1) Datasets: We evaluate our method on three public datasets to validate its applicability to both IoT and traditional networks. Notably, these datasets were collected in 2019, 2020, and 2024 respectively, demonstrating that our approach can adapt to evolving network environments.

- *CIC-DDoS2019 [74]:* Designed for DDoS attack detection, this dataset introduces a new taxonomy for DDoS attacks. It contains benign traffic and 12 up-to-date common DDoS attacks, such as NetBIOS, LDAP, and MSSQL.
- *MQTT-IoT-IDS2020 [75]:* Focused on the MQTT protocol, a core IoT communication protocol, this dataset simulates real-world network environments. It covers normal operations and four attack scenarios, enabling validation of our method's detection capability in IoT environments.
- *CICIoMT2024 [76]:* Targeting Medical IoT (IoMT) scenarios, this dataset contains five categories of network attacks against 40 IoMT devices, spanning Wi-Fi, MQTT, and Bluetooth protocols. Combining real medical devices with simulated attacks, it validates our method's effectiveness in real-world complex IoT environments.

For all datasets, network flows are divided based on five-tuples. Given the significant class imbalance in CIC-DDoS2019 and CICIoMT2024, we balance the datasets by randomly sampling other major categories using benign traffic as the baseline. Each dataset is divided into training and test sets at a 4:1 ratio. Detailed sample distributions are shown in Table III.

2) Experimental Setup: We train and evaluate the model using PyTorch on a server equipped with an NVIDIA RTX 3090 GPU. The PyTorch Geometric Library is employed to construct the ISTGs and implement the GNN. The relevant parameters of the model and training are listed in Table IV. To make full use of the training data, 5-fold cross-validation is adopted during the training process.

3) Evaluation Metrics: To assess the classification performance of the model, the following four well-known metrics are adopted: Accuracy (ACC), Precision (Positive Predictive Value, PPV), Recall (True Positive Rate, TPR), and F1 Score (F1). The F1 Score provides a comprehensive measure of model performance, representing the harmonic mean of PPV and TPR. To complement this assessment with insights into efficiency and practicality, we also measure the model's parameter count, training/inference time, memory usage, and throughput.

TABLE III
The Composition of the Three Dataset

Dataset	Category	Flow count		Dataset	Category	Flow count	
		Train	Test			Train	Test
[36]	Benign	64000	16000	[37]	Benign	70560	17640
	DNS	11688	2928		scan_A	20688	5176
	LDAP	64000	16000		scan_sU	32096	8024
	MSSQL	64000	16000		bruteforce	13920	3480
	NetBIOS	64000	16000		sparta	74576	18648
	SSDP	64000	16000				
	SYN	64000	16000	[38]	Benign	8000	2000
	UDP	64000	16000		Spoofing	632	160
	WebDDoS	968	240		MQTT	8000	2000
	TFTP	64000	16000		DDoS	8000	2000
	SNMP	64000	16000		DoS	8000	2000
	NTP	64000	16000		Recon	8000	2000
	UDP-Lag	3504	880				

TABLE IV
Parameter Setting

Parameter	Meaning	Value
m	The number of nodes contained in a graph	12
q	The truncation length for byte sequence	300
p	The truncation length for length sequence	20
z	The number of statistical features	6
F_{CNN}	The hidden vector dimension of CNN	1024
F_{GNN}	The hidden vector dimension of GNN	128
bs	Batch size	128
lr	Learning rate	0.0005
$optimizer$	Optimizer	Adam
$epochs$	Training epochs	100
C	The number of network flow categories	13(DDoS2019) 5(MQTT-IoT2020) 6(CICIoMT2024)

4) Statistical Feature Selection: Node features in the ISTG consist of packet length sequences and statistical features. The statistical features capture the overall characteristics of the flows, serving as a supplement to the truncated sequence features. As described in Section. IV-C1, we compute 18 statistical features for each of the entire, forward, and backward packet length sequences, resulting in a total of 54 statistical features. The Fig. 4 shows the top 10 most important statistical features ranked by their contribution using a random forest classifier, where f, b, and a represent the forward, backward, and entire flows, respectively. To reduce the model parameters while maintaining learning accuracy, we select the six most significant statistical features: a_mean, f_per4, a_per7, f_per5, a_per5, and a_per6. Consequently, the feature dimension of each node in the graph is $p + z = 26$.

B. Comparison With State-of-The-Art Methods

From the perspective of feature utilization, the following comparative methods are selected: Fs-Net, 1D-CNN, and XGBoosts employ single-modal features of packet length sequences, raw bytes, and statistical features, respectively, while both APPNet

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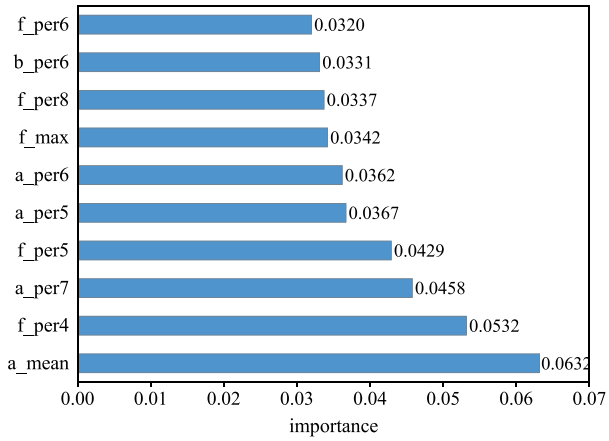


Fig. 4. Top 10 most important statistical features ranked using random forest.

and iDetector adopt multimodal features. All these five models focus exclusively on intra-flow features. In contrast, GAT-based model, PPT-GNN, and CCSTGN explore inter-flow correlations through graph construction. MeDF and IIT are conceptually similar to ours, as both attempt to combine intra-flow and inter-flow features.

- Fs-Net [28] employs a bidirectional GRU to encode features from packet length sequences and introduces a reconstruction mechanism in the AutoEncoder to ensure the effectiveness of the learned features.
- 1D-CNN [31] converts sequences of hexadecimal bytes from encrypted network flows into grayscale images and utilizes a 1D CNN model to classify these images.
- XGBoost [19] extracts statistical features related to time intervals and packet sizes, and utilize the XGBoost algorithm for classification.
- App-Net [48] extracts features from byte sequences using a 1D CNN and learns temporal relationships in packet length sequences using LSTM. Finally, the features learned from both views are concatenated for classification.
- iDetector [41] extracts three sequence features of byte, length and time interval from channel traffic, and classifies them using a lightweight deep neural network model called EdgeNet.
- GAT_based model [52] treats network flows as nodes, embedding 63 statistical features including packet size, packet count, protocol type, and IP addresses into each node. Edges are added based on cross-correlations between nodes, and a GAT is applied to classify the graph.
- PPT-GNN [65] constructs a heterogeneous graph with flows and IP addresses as nodes, and employs multiple heterogeneous spatio-temporal layers to capture spatial, temporal, intra-window, and inter-window dependencies.
- CCSTGN [66] maps flows to communication channels, constructing a channel-centric graph representation. It utilizes a Temporal Graph Network for continuous-time dynamic graph learning, thereby capturing complex spatio-temporal dependencies among traffic flows.
- IIT [63] concatenates packet length sequences, time interval sequences, and length cumulative sum sequences as input features. Two multi-head self-attention-based encoders are leveraged to mine intra-flow and inter-flow relationship.
- MeDF [54] combines intra-flow and inter-flow features: it constructs spectrograms of flows' raw bytes and extracts statistical features, combining the two as the intra-flow features. MeDF also builds flow relation graph to extracts inter-flow features. For consistency, we similarly construct graphs by grouping adjacent m flows together.

Table V compares the classification performance of each model across three datasets. For a fair comparison, all baseline methods were re-implemented and evaluated on the same balanced datasets under consistent experimental settings. The experimental results demonstrate that our proposed IF^2-STG achieves a classification accuracy of over 99% on all three datasets, outperforming other comparative models, which indicates its good generalization capability across different network environments. Particularly, its performance on the CIC-DDoS2019 dataset confirms that IF^2-STG can not only effectively distinguish between normal and attack traffic, but also precisely identify different attack subtypes.

Among the comparative models, Fs-Net, which utilizes packet length sequences, shows the worst performance, achieving barely over 60% accuracy on the first two datasets. This indicates that coarse-grained length sequence features contain insufficient information to effectively distinguish malicious network behaviors. Similarly, the XGBoost model based on statistical features exhibits limited generalization. In contrast, the single-modal 1D-CNN approach, using only raw byte features, achieves better results. Notably, on the CIC-DDoS2019 dataset, all models achieving an accuracy of over 90% incorporate byte features, underscoring that such fine-grained data is essential for the detailed classification of specific attack subcategories.

The multi-modal models AppNet and iDetector demonstrate superior performance compared to the aforementioned three single-modal models, with classification accuracies both reaching around 97% . This indicates that multi-modal approaches can effectively integrate diverse features to more comprehensively characterize network flow patterns.

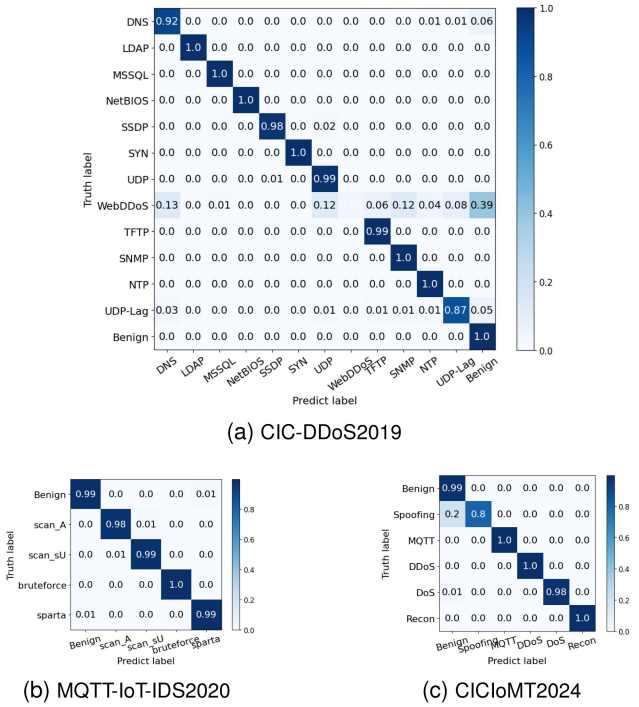
The GAT_based model adds edges based on the cross-correlation between nodes, which captures only the temporal dependencies between flows while ignoring spatial structural features. Both PPT-GNN and CCSTGN dynamically capture spatio-temporal interactions among flows. However, as they rely solely on statistical features and omit fine-grained intra-flow details, their performance shows obvious instability across the three datasets. By contrast, our IF^2-STG method achieves superior and more robust detection performance through joint modeling of inter-flow spatio-temporal features and intra-flow fine-grained features.

The IIT method employs a complex attention mechanism to implicitly mine intra-flow and inter-flow features and has the highest parameter count. However, its classification performance is unsatisfactory, primarily because it relies mainly on packet length sequences. Nevertheless, thanks to the incorporation of time interval features, its performance is slightly better than Fs-Net, especially on the CICIoMT2024 dataset.

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TABLE V
CLASSIFICATION RESULTS OF ALL METHODS ON THE THREE DATASETS

Model	Params	CIC-DDoS2019				MQTT-IoT-IDS2020				CICIoMT2024			
		ACC(%)	PPV(%)	TPR(%)	F1(%)	ACC(%)	PPV(%)	TPR(%)	F1(%)	ACC(%)	PPV (%)	TPR(%)	F1(%)
Fs-Net	3.35 M	62.15	65.40	62.15	57.16	67.07	83.58	67.07	65.31	79.54	79.07	79.54	72.84
1D-CNN	5.82 M	93.36	94.81	93.36	92.91	97.36	97.42	97.36	97.37	95.94	96.42	95.94	95.70
XGBoost	–	76.60	77.83	76.60	75.86	80.21	80.62	80.21	80.29	93.63	93.62	93.63	93.61
AppNet	2.30 M	97.75	98.14	97.75	97.90	97.57	97.57	97.57	97.56	98.09	98.12	98.09	98.02
iDetector	1.23 M	96.71	96.80	96.71	96.70	97.45	97.54	97.45	97.46	97.59	97.53	97.59	97.42
GAT_based model	0.24 M	87.68	88.17	87.68	87.43	82.98	83.42	82.98	83.11	96.44	96.45	96.44	96.44
PPT-GNN	0.47 M	86.33	82.83	86.33	83.50	82.70	87.69	82.70	80.01	94.73	95.25	94.73	94.70
CCSTGN	0.01 M	83.68	77.97	83.68	79.55	96.73	96.90	96.73	96.64	91.30	88.89	91.30	89.34
IIT	11.3 M	62.46	67.41	62.46	59.06	72.26	75.05	72.26	72.76	89.04	88.99	89.04	88.91
MeDF	0.32 M	98.11	98.33	98.11	98.18	97.96	97.96	97.96	97.96	98.15	98.11	98.15	98.07
IF^2-STG	3.28 M	**99.10**	**99.04**	**99.10**	**99.03**	**99.27**	**99.27**	**99.27**	**99.27**	**99.58**	**99.59**	**99.58**	**99.59**



(a) CIC-DDoS2019

(b) MQTT-IoT-IDS2020

(c) CICIoMT2024

Fig. 5. Confusion matrices of IF^2-STG on three datasets.

MeDF achieves the best performance among the comparative methods, with an accuracy of approximately 98% on all three datasets, verifying the effectiveness of combining intra-flow and inter-flow features. However, its graph construction, which considers only spatial correlations, is limited. Additionally, the processing method of converting raw bytes into time-frequency spectrograms may risk information loss. Ablation experiments in Section. V-D will verify whether this feature transformation approach proves more effective than directly using raw byte sequences.

Fig. 5 presents the confusion matrices of our method across the three datasets. It can be observed that most categories achieve a prediction accuracy close to 1. The relatively lower accuracy in certain categories stems from insufficient training samples. For instance, the WebDDoS category in the CIC-DDoS2019 dataset only has 968 training samples, significantly fewer than other categories.

TABLE VI
CLASSIFICATION PERFORMANCE ON IMBALANCED CICIOMT2024 DATASET

Category	Count	PPV(%)	TPR(%)	F1(%)
Benign	10884	96.20	99.13	97.64
Spoofing	792	91.43	82.05	86.49
MQTT	14196	99.86	99.93	99.89
DDoS	20748	99.81	99.86	99.83
DoS	14052	99.39	98.43	98.91
Recon	128328	99.96	99.85	99.90
Macro Avg	-	97.77	96.54	97.11
Weighted Avg	-	99.64	99.64	99.64
Accuracy	-		99.64	

Furthermore, Fig. 6 compares the time costs of all models, including training time, inference time, and sample construction time. In terms of training and inference efficiency, the proposed IF^2-STG model demonstrates competitive performance, achieving comparatively fast speeds. iDetector, Fs-Net, and PPT-GNN are the three models with the highest time overhead. For sample construction time, graph-based methods generally require more time than those based on individual flows. Among them, PPT-GNN takes the longest, as it needs to construct all traffic within a time window into a single large graph. For short-term burst attacks like DDoS, even with a small time window (e.g., 0.5 seconds), the resulting graph may contain tens of thousands of nodes, leading to a slow graph-building process and further reducing its training and inference efficiency.

C. Evaluation on Imbalanced Data

Data imbalance is a prevalent and critical issue in real-world scenarios. In this section, the proposed method is trained and tested on the original, unbalanced CICIoMT2024 dataset. This dataset exhibits an extreme long-tail distribution (see Table VI), where the sample ratio between the majority class (Recon) and the minority class (Spoofing) reaches 162:1. This distribution realistically reflects the challenge of detecting rare attack types in actual network traffic environments.

As summarized in Table VI, the proposed method demonstrates strong overall classification performance under such highly imbalanced conditions. Notably, it retains considerable detection accuracy for the minority class (Spoofing), indicating that the model does not exhibit significant learning bias toward

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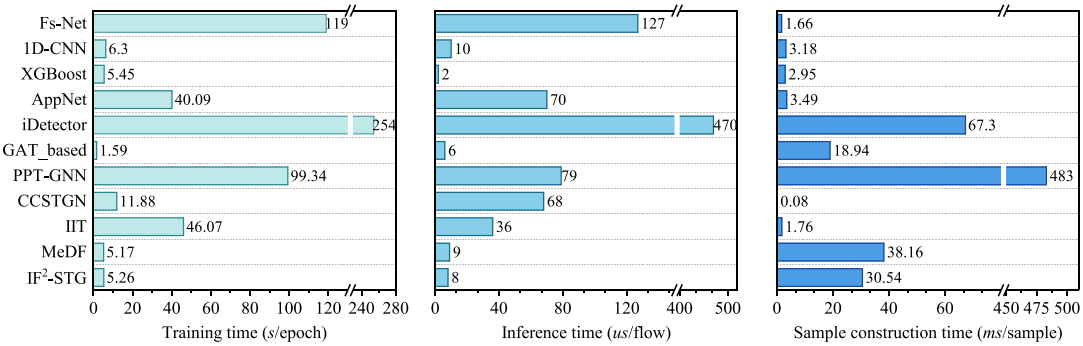


Fig. 6. Comparison of time cost of each baseline model.

the majority class. These results validate the method's effectiveness in real-world scenarios characterized by data imbalance.

D. Ablation Experiment

In this section, we construct three ablated models to further validate the effectiveness of combining inter-flow and intra-flow features, as well as the multi-loss joint training approach. Additionally, to investigate whether MeDF's Fourier transform on raw bytes to obtain time-frequency spectrograms is more effective, we introduce an ablation experiment MeDF-Intra that adopts MeDF's processing method for raw bytes and the learning network. The relevant parameters of the short-time Fourier transform are kept consistent with the original MeDF, such as a window length of 100, an overlap rate of 67% , and a Hanning window function. Thus, the four sets of ablation experiments are as follows. We perform experiments on the MQTT-IoT-IDS2020 dataset.

- *IF²-STG-Intra:* Removes the inter-flow feature extraction module from IF²-STG, retaining only the intra-flow feature extraction module.
- *IF²-STG-Inter:* Removes the intra-flow feature extraction module from IF²-STG, retaining only the inter-flow feature extraction module.
- *MeDF-Intra:* Adopts MeDF's processing method for raw bytes and the learning network.
- *IF²-STG-loss:* Uses only the single loss from the final fusion classification module for backpropagation optimization.

As shown in Fig. 7, the joint learning scheme integrating both intra-flow and inter-flow features significantly enhances intrusion detection performance. Compared to the ablated models using only intra-flow features or only inter-flow features, the complete IF²-STG model achieves accuracy improvements of 1.67% and 2% respectively. This demonstrates that the intra-flow and inter-flow features we employ exhibit complementary advantages during modal fusion, thereby enhancing overall performance. Specifically, the intra-flow features derived from raw bytes capture fine-grained protocol semantics and packet-level patterns, while the inter-flow features model spatio-temporal correlations between flows. The two modal features complement each other, overcoming the representational limitations

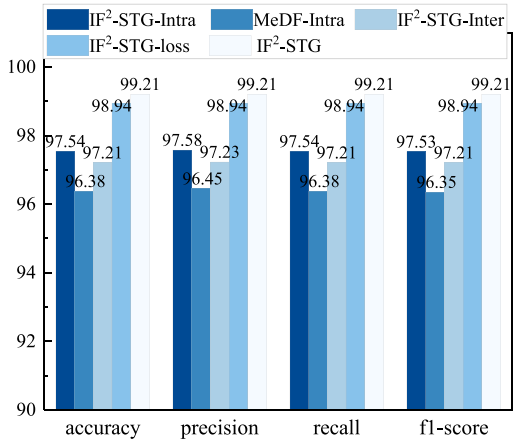


Fig. 7. Ablation study results.

of single-modal approaches and enabling more comprehensive analysis of network behavior.

Furthermore, MeDF-Intra exhibits approximately 1% lower performance than IF²-STG-Intra across four classification metrics. This observation suggests that converting raw byte sequences into time-frequency spectrograms does not enhance feature discriminability, but may instead degrade classification performance. The reason lies in that the raw byte sequences completely preserve the structured characteristics of network flows, such as protocol header fields, which can be directly captured by the classification model. In contrast, during the Fourier transform and spectrogram generation process, operations like window truncation may damage original features, cause information loss, or introduce noise. These factors make it difficult for spectrograms to present effective discriminative features, ultimately leading to decreased classification performance.

Additionally, IF²-STG achieves better performance than IF²-STG-loss, demonstrating that employing multi-loss joint training can help the model better balance the relationship between intra-flow and inter-flow modal features, thereby bringing performance gains.

E. Sensitivity Analysis

This section analyzes the impact of key parameters on the overall performance of the model: the intra-flow feature

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TABLE VII
THE IMPACT OF m ON CLASSIFICATION PERFORMANCE AND EFFICIENCY

m	ACC(%)	F1(%)	Training time (s/epoch)	Memory usage (MB)	Inference time (us/graph)	Throughput (flows/s)
4	98.67	98.67	7.89	2360	77.42	51664
8	98.98	98.98	4.76	2540	86.36	92630
12	**99.27**	**99.27**	4.69	2684	99.59	120499
16	99.09	99.09	3.24	2842	107.85	148355
20	99.07	99.07	2.88	3018	111.65	179123

TABLE VIII
THE IMPACT OF q ON CLASSIFICATION PERFORMANCE AND EFFICIENCY

q	ACC	F1	Params	Training time (s/epoch)	Memory usage (MB)	Inference time (us/graph)	Throughput (flows/s)
100	98.75	98.75	1.83 M	2.77	2400	89.24	134473
300	**99.27**	**99.27**	3.28 M	4.69	2684	99.59	120499
500	99.21	99.21	4.73 M	5.62	2966	119.94	96741
1000	99.04	99.04	8.39 M	7.01	3688	148.91	80583
2000	99.09	99.09	15.66 M	10.98	4670	185.84	64569

extraction parameter q and the inter-flow feature extraction parameter m. In addition, it investigates the effect of two critical designs in the construction of the ISTG—the time threshold for partitioning concurrent groups and the edge definition rules—on the detection performance of the IF^2-STG-Inter model.

1) Number of Nodes m in ISTG: We conducted experiments with m = 4, 8, 12, 16 and 20, respectively, and the results are listed in Table VII. In terms of detection performance, both classification accuracy and the F1-score show a trend of first increasing and then decreasing as m grows, reaching the optimal values at m=12. In the first phase, as m increases, each ISTG incorporates more flows, enabling the model to capture richer inter-flow interaction patterns, thereby improving classification accuracy. However, when m continues to grow beyond 12, the classification performance deteriorates. This phenomenon occurs because the total number of network flows remains constant. Larger values of m lead to fewer graph samples, which in turn restricts the model's learning ability. Moreover, it is clear that larger m values result in longer training times, higher GPU memory usage, and greater demands for computational resources. In light of these findings, we set m=12 as the default configuration.

During the inference phase, the classification results for individual flows can only be obtained after the entire graph has been processed. Thus, the inference latency is determined by the average detection time per graph. The results show that as m increases, the processing time per graph becomes longer, leading to higher system response latency. However, since each graph contains more flows, the throughput is improved instead. This reflects the trade-off between latency and throughput with respect to graph scale. In practical deployment, the value of m can be chosen according to specific requirements.

2) Length of the Byte Sequence q: We conducted experiments with q = 100, 300, 500, 1000, and 2000, respectively, and the results are summarized in Table VIII. Experimental results indicate that the model achieves optimal classification performance when the first 300 bytes of the flow are intercepted. Further increasing the sequence length leads to a degradation in performance, suggesting that longer byte sequences do not contribute additional discriminative information and may instead introduce noise that interferes with the model's decision-making.

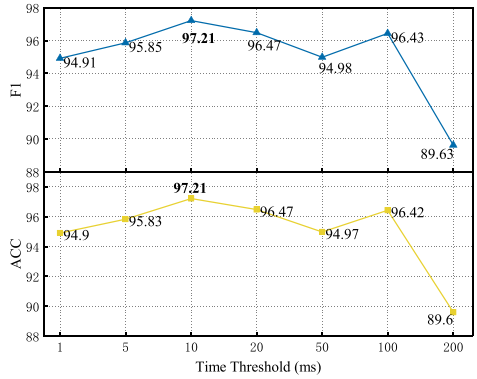


Fig. 8. The Impact of Time Threshold on Classification Performance.

Moreover, all efficiency metrics show a strong positive correlation with q, as it directly determines the input dimension and computational complexity of the model. Particularly, the inference throughput decreases sharply as q grows. It drops by 46% when q goes from 300 to 2000. This represents a critical trade-off in large-scale network traffic analysis scenarios. Consequently, q = 300 is selected as the default configuration to strike a balance between classification accuracy and computational efficiency.

3) Time Threshold: Flows with a time difference smaller than a specific threshold and an intersection between the destination IP and source IP sets are considered concurrent. This time threshold directly determines the size of concurrent groups, thereby influencing the constructed graph structure and the effectiveness of inter-flow feature learning. To evaluate its impact, we tested the performance of the IF^2 -STG-Inter model across threshold values ranging from 1ms to 200ms. The results, shown in Fig. 8, indicate that the model performs best at a threshold of 10ms. If the threshold is excessively low, concurrent groups may contain too few flows to supply sufficient contextual information. Conversely, an overly high threshold will incorporate temporally weakly correlated flows into the same group, thereby introducing noise. Thus, a 10ms threshold effectively captures meaningful concurrent flow behavior, representing a balanced and robust configuration.

4) Edge Definition Rules: This paper selects start time intervals and IP-set associations as the core criteria for constructing spatio-temporal edges in the ISTG, primarily based on two considerations: First, multi-flow coordination in network attacks (e.g., DDoS) often manifests as temporally dense triggering and IP-level distributed associations. These dual-dimensional features can accurately characterize the spatio-temporal collaboration patterns of such attacks. Second, building upon existing research—notably the MeDF model [54] that uses IP-set associations to model spatial flow relationships—this paper further introduces the dimension of time intervals, thereby enabling complementary modeling of spatio-temporal features. To verify

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TABLE IX
THE IMPACT OF EDGE DEFINITION RULES ON CLASSIFICATION PERFORMANCE

Rule	ACC(%)	PPV(%)	TPR(%)	F1(%)
Time Interval Only	95.51	95.57	95.51	95.52
IP-set Association Only	86.50	86.93	86.50	86.61
Time Co-occurrence + IP-set Association	87.68	88.00	87.68	87.79
Time Interval + Source IP Match	96.01	96.12	96.01	96.03
Time Interval + Destination IP Match	96.33	96.36	96.33	96.33
Time Interval + Port-set Association	96.04	96.08	96.04	96.04
Time Interval + IP-set Association	**97.21**	**97.23**	**97.21**	**97.21**

the rationality of the two selected indicators, we compared the following six edge definition rules:

- Time Interval Only: Constructs one-dimensional edges based solely on time intervals, ignoring spatial associations.
- IP-set Association Only: Constructs one-dimensional edges based solely on IP-set associations, ignoring temporal associations.
- Time Co-occurrence + IP-set Association: Temporal association follows the time co-occurrence rule of the GAT_based model [52], where two flows overlapping in time are considered related and assigned a weight of 1.0.
- Time Interval + Source IP Match: Spatial association is simplified to matching only whether the source IPs are identical.
- Time Interval + Destination IP Match: Spatial association is simplified to matching only whether the destination IPs are identical.
- Time Interval + Port-set Association: Spatial association is replaced by whether there is an intersection in the port sets.

Table IX compares the model's performance under different edge definition rules, leading to the following conclusions. First, the proposed "time interval + IP-set association" rule achieves the best performance. Second, single-dimensional rules (time interval only or IP-set association only) deliver inferior performance to dual-dimensional rules, with the time interval contributing more critically. Third, the "time co-occurrence + IP-set association" rule performs notably worse than the rule we adopt, since time intervals more precisely capture concurrent patterns among multiple flows compared to time co-occurrence. Fourth, IP-set association outperforms single IP matching because it can reveal complex attack topologies involving many-to-many and chain-style communications, thereby demonstrating its stronger generalization capability in spatial modeling. Fifth, replacing spatial association with port-set associations results in a slight performance decline, reflecting that port features are less stable and discriminative than IP addresses. For example, in port scanning attacks, malicious flows exhibit clear correlations at the IP level, whereas port numbers often change randomly or dynamically, failing to provide stable correlation signals.

In summary, the "time interval + IP-set association" rule offers considerable benefits for ISTG construction. It effectively captures the temporal density of attack flows while enabling a precise representation of their spatial logic, thereby furnishing a more discriminative graph-structured input for downstream detection models.

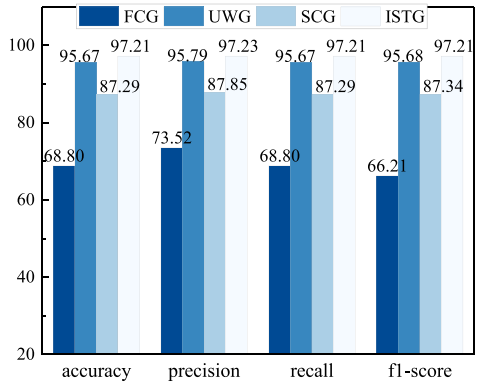


Fig. 9. The Impact of ISTG Construction Scheme on Classification Performance.

F. Impact of ISTG Construction Scheme

The construction of ISTG forms the foundation and key to inter-flow feature learning. Therefore, this section analyzes the impact of two-dimensional spatio-temporal vector edges and concurrent groups in our proposed ISTG on the classification performance of the inter-flow feature module, through the following three comparative experiments:

- *Fully-connected Graph (FCG):* All node pairs in the graph are connected with uniform edge weights set to 1.
- *Uniform-weight Graph (UWG):* Maintaining the original ISTG topology while replacing all edge weights from two-dimensional spatio-temporal vectors with 1.
- *Sequential-connection Graph (SCG):* All nodes are connected sequentially in chronological order, disregarding concurrent flows.

As depicted in Fig. 9, the FCG performs the worst, with an F1 score of 66.21% . In this case, all graphs share the same topology and edge weights, differing solely in node features, which fails entirely to reflect the differences in interaction patterns among network flows of different categories. In comparison, the UWG preserves the original ISTG topology while simplifying edge weights to 1, yet achieves substantially better results. This empirically validates the intrinsic discriminative ability of topological structures in inter-flow relationship learning. Nevertheless, compared with ISTG utilizing two-dimensional spatio-temporal vectors, UWG's F1 score drops by 1.53% . This gap highlights that two-dimensional edge weights facilitate more precise spatio-temporal correlation modeling, thereby enhancing the model's discrimination capacity in complex traffic scenarios.

The SCG model without concurrent flow consideration exhibits a notable decline in detection performance (F1=87.34%). This comparative result demonstrates the effectiveness of our designed graph construction scheme that considers concurrent flows, which is better at capturing interaction frequencies between network flows. Network attacks and anomalous behaviors frequently involve special concurrent patterns. For example, in distributed denial-of-service (DDoS) attacks, attackers typically

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TABLE X
DIVISION OF ATTACK TYPES FOR THE THREE DATASETS

Dataset	DDoS/DoS attacks	scanning/bruteforce attacks
CIC-DDoS2019	DNS, LDAP, MSSQL,NetBIOS, SSDP, SYN, UDP, WebDDoS, TFTP, SNMP, NTP, UDP-Lag	–
MQTT-IoT-IDS2020	–	scan_A, scan_sU, bruteforce, sparta
CICIoMT2024	DDoS, DoS, MQTT-DoS-Connect_Flood, MQTT-DDoS-Connect_Flood, MQTT-DoS-Publish_Flood, MQTT-DDoS-Publish_Flood	Recon-Ping_Sweep, Recon-Port_Scan, Recon-OS_Scan, Recon-Vulnerability_Scan

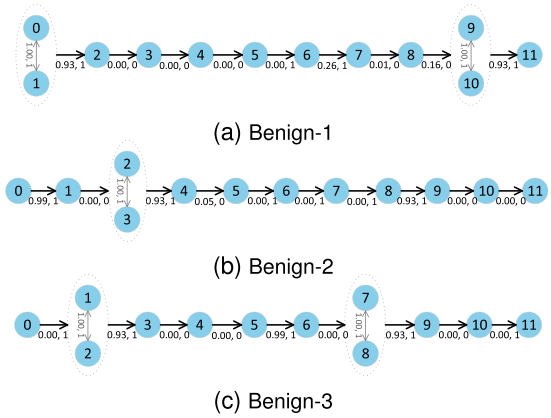


Fig. 10. ISTG corresponding to benign traffic.

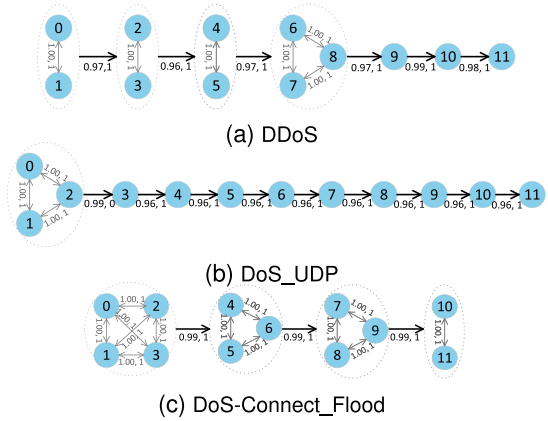


Fig. 11. ISTG corresponding to DDoS/DoS traffic.

initiate massive concurrent requests from multiple source IPs toward the target within a short period. By grouping flows occurring within 10 ms intervals for graph construction, our method amplifies these abnormal concurrency characteristics, thereby allowing the model to more effectively differentiate between normal and malicious traffic.

G. Visualization Analysis of ISTG

To further verify the effectiveness of the ISTG construction method, we conduct a comparative visual analysis of ISTG generated from different categories of network flows. As shown in the Table X, according to attack characteristics, the malicious traffic in the three datasets used in this paper can be mainly divided into two major categories: DDoS/DoS attacks and scanning/bruteforce attacks. Among them, those prefixed with "MQTT-" or "Recon-" represent subcategories under the corresponding major categories.

We randomly select three instances from benign traffic and the two malicious traffic categories, with their corresponding ISTGs visualized in Figs. 10–12. It can be seen that the ISTGs of different types of flows exhibit obvious differences, conclusively validating the effectiveness of our proposed ISTG construction strategy. The rationality analysis of the characteristics of the three ISTG types is carried out below in combination with the inherent characteristics of network flows.

In the ISTGs corresponding to benign traffic (Fig. 10(a)–(c)), nodes basically distributed in a chain-like manner, and there are relatively few concurrent groups. The temporal components t of unidirectional edges are predominantly small, indicating

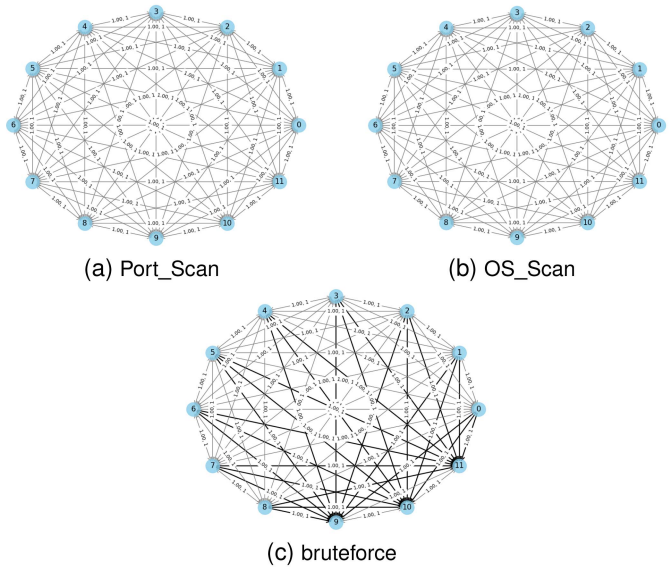


Fig. 12. ISTG corresponding to scanning/bruteforce traffic.

relatively large inter-flow time intervals, while spatial components s display both 1 and 0 values. This pattern arises because benign traffic typically involves normal network service interactions, featuring few concurrent scenarios, gentle time distribution, and random IP interactions without fixed patterns.

In the ISTGs corresponding to DDoS/DoS attacks (Fig. 11(a)–(c)), there are more concurrent flows. Moreover, for the unidirectional edges, the time component t is nearly 1, and the spatial component s is almost all 1. This is

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because such attacks require sending traffic to the target in a short time (or even concurrently) in terms of time, and the targets are concentrated in space, showing spatial aggregation. Additionally, the key difference between the ISTGs of the two DoS attacks (b) and (c) mainly lies in the number of concurrent groups, which is rooted in the attack protocols used. The DoS_UDP attack exploits UDP's connectionless, stateless properties to flood random target ports with UDP packets. The DoS-Connect_Flood belongs to TCP connection exhaustion attacks (such as SYN Flood and HTTP Connect Flood), which exhaust target resources by quickly establishing numerous TCP connections. The stateful connection mechanism of TCP compels attackers to maintain high-concurrency traffic to enhance the degree of damage. Therefore, the latter generating more concurrent groups than the former.

The ISTGs corresponding to scanning/bruteforce attacks (Fig. 12(a)–(c)) exhibit fully-connected mesh topologies. It is noticeable that all 12 nodes in each graph are grouped into one or two concurrent groups. This pattern emerges because both attack types typically send probing or cracking requests within extremely short time windows. They are highly concurrent in terms of time, and the IP addresses of the attacking hosts or the target hosts are fixed. Therefore, they meet the requirements of concurrent flows.

H. Cross-Dataset Generalization

In this section, we integrate three datasets at different times to evaluate the ability of IF^2-STG and comparative methods to identify known attacks in a cross-dataset environment. We merge the CIC-DDoS2019 and MQTT-IoT-IDS2020 datasets to train a binary classification model, which is then tested on the known attacks (DDoS and Recon) from the CICIoMT2024 dataset. Furthermore, recognizing that benign traffic may exhibit different characteristics across datasets collected at different time periods and is easy to collect in real-world scenarios, we incorporate part of the benign traffic from the CICIoMT2024 dataset into the combined dataset for training. In practice, new normal traffic can be collected and added to update the model.

The detection results and the features used by the models are shown in Table XI. To explore the generalization ability of inter-flow features, we specifically tested variants of the MeDF and IF^2-STG models that utilize inter-flow features exclusively. The experimental results reveal significant differences in different feature types under cross-dataset scenarios. Our IF^2-STG-Inter achieves 97.78% accuracy, recall, and F1-score, indicating that inter-flow spatio-temporal relationships effectively capture the essential behavioral patterns of attacks, including communication frequency, topological connections, etc. These features maintain remarkable stability across datasets collected at different times. However, the complete IF^2-STG model incorporating intra-flow raw byte features attains only 42.91% accuracy, as raw bytes are excessively dependent on network environments, protocol versions, and encryption methods, etc., severely limiting model transferability. Other comparative models using raw byte features exhibit similar limitations: 1D-CNN, AppNet, and MeDF achieve merely 33.83% , 58.08% , and 36.39% recall rates respectively, meaning that substantial numbers of attacks being misclassified as benign traffic, posing a serious risk of false negatives in real-world security scenarios.

TABLE XI
TEST RESULTS ON CROSS-DATASET GENERALIZATION

Model	Intra-feature				Inter-feature		ACC	PRE	REC	F1
	RBS	PLS	IATS	STA	TR	SR				
Fs-Net		✓					85.55	88.06	85.55	84.27
1D-CNN	✓						33.83	72.27	33.83	17.81
XGBoost				✓			67.65	82.92	67.65	67.82
AppNet	✓	✓					58.08	81.39	58.08	56.57
iDetector	✓	✓	✓				78.96	86.31	78.96	79.05
GAT_based				✓	✓		61.25	70.67	61.25	61.87
PPT-GNN				✓	✓	✓	75.97	76.53	75.97	74.76
CCSTGN				✓	✓	✓	88.44	89.80	88.44	83.98
IIT		✓	✓		✓	✓	66.54	68.69	69.06	66.52
MeDF	✓			✓		✓	36.39	77.42	36.39	22.96
MeDF-Inter						✓	33.33	11.11	33.33	16.67
IF^2-STG	✓				✓	✓	42.91	78.27	42.91	34.82
IF^2-STG-Inter					✓	✓	**97.78**	**97.81**	**97.78**	**97.78**

RBS: raw byte sequences; PLS: packet length sequences; IATS: inter-arrival time sequences; STA: statistical features; TR: temporal relationship; SR: spatial relationship;

It is noteworthy that iDetector, despite also utilizing raw byte features, still achieves an accuracy of 78.96% . This may benefit from its incorporation of inter-arrival time sequences and packet length sequences, which more effectively capture the temporal distribution patterns of attack behaviors. Additionally, Fs-Net, XGBoost and IIT models perform relatively well, with Fs-Net in particular attaining approximately 85% accuracy and F1-score. They employ packet length sequences and statistical features, which are more abstract and coarse-grained compared to raw bytes. Such features rely less on traffic details in specific datasets and can, to a certain degree, better reflect the intrinsic characteristics of attack behaviors.

Additionally, among the comparative methods, GAT_based model and MeDF-Inter, although also based on inter-flow features, only extract temporal or spatial relationships between flows without joint modeling of spatio-temporal dependencies. This results in insufficient cross-dataset generalization ability, with their accuracy rates being only 61.25% and 33.33% respectively. Further analysis reveals the critical impact of node feature selection in graph representation learning. GAT_based model employs length statistics and IP addresses as node features, whereas MeDF-Inter uses only IP addresses and protocol numbers as node features. The latter's insufficient feature information leads to weaker representational ability than the former. PPT-GNN and CSSTGN both use statistical features and model the spatio-temporal relationship among flows, so the transferability is slightly better.

In contrast, our IF^2-STG-Inter adopts packet length sequences and statistical features as node features. As previously analyzed, these two types of features demonstrate better generalization across different datasets. Moreover, by jointly modeling both temporal and spatial relationships between flows, our approach achieves a more comprehensive characterization of attack behaviors. Consequently, it shows significant advantages in cross-dataset detection, offering a highly adaptable security

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TABLE XII
DETECTION PERFORMANCE OF THE MODEL ON MIXED TRAFFIC

Model	ACC(%)	PPV(%)	TPR(%)	F1(%)
IF^2-STG	100	100	100	100
IF^2-STG-Inter	99.65	99.65	99.65	99.65

detection solution for dynamically evolving attacks and network environments.

I. Adaptability to Mixed Traffic

The graph construction strategy in this work is based on the assumption that m adjacent flows belong to the same category. However, in real-world network detection environments, this assumption cannot always be guaranteed in advance. Therefore, this section aims to validate the detection performance of the proposed method when facing realistic mixed traffic.

For the data selection, this study employs the CICIDS2017 dataset [77]. The original PCAP files provided in this dataset contain unsegmented full-day traffic, which more closely reflects real network environments. According to the official description, on the afternoon of Friday, July 7, 2017, port scan attacks were launched at one-minute intervals, each lasting approximately 2–3 minutes, against a background of normal communication traffic. We intercept the traffic during this specific period to simulate a scenario where attack traffic is interleaved with normal business traffic.

After processing and constructing graphs from this mixed traffic, we obtain three types of graphs: 5,626 graphs containing only benign nodes, 13,120 graphs consisting solely of attack nodes, and 431 mixed graphs containing both benign and attack nodes. It should be noted that due to the presence of multiple node labels in mixed graphs, we adjust the task from graph classification to node classification. Specifically, in the inter-flow feature extraction module (Fig. 2), instead of using SAGPool for graph pooling after the GATConv layer, we directly connect a linear layer to output node-level classification results.

To evaluate the learning effectiveness of the spatio-temporal relation graph, in addition to the complete model, we also test the IF^2-STG-Inter model. The experimental results are shown in Table XII. In this binary classification task, the complete model achieves 100% accuracy. The IF^2-STG-Inter model also reaches an accuracy of 99.65% , strongly demonstrating the exceptional discriminative capability of our constructed ISTG even in mixed traffic scenarios.

In addition, we visualize the ISTG containing mixed-type nodes, as shown in Fig. 13. In the figure, the marker on the outer ring of each node represents its corresponding label. For clarity, normal traffic is marked in green, and port scan attacks are indicated in red. In subfigure (a), a single normal flow embedded among multiple attack flows appears as an isolated node, while the remaining nodes form a dense concurrent cluster. In subfigure (b), the attack flows at the beginning and the end form their own concurrent clusters, with benign flows in the middle connected in a linear pattern. Moreover, the spatial component attribute of edges between adjacent benign and attack nodes has a value of 0, indicating no spatial correlation between them.

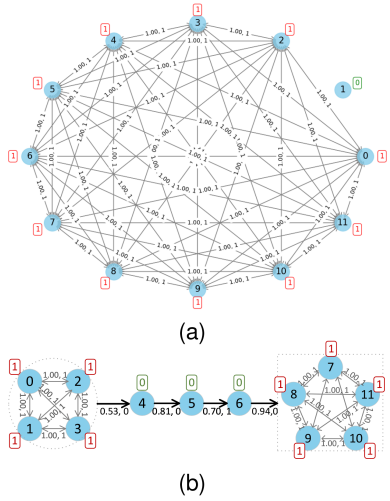


Fig. 13. ISTG corresponding to mixed traffic.

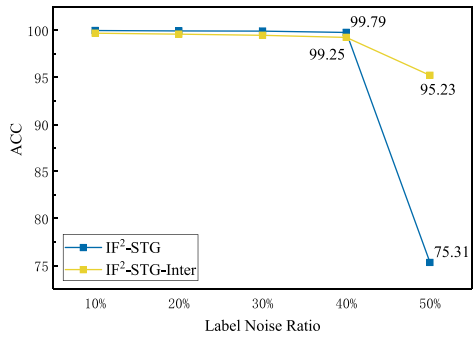


Fig. 14. The Impact of labeling noise on Classification Accuracy.

Furthermore, to evaluate the model's robustness to labeling noise, we introduce random label noise into the training set. The classification accuracy of the two models at different noise levels is depicted in Fig. 14. Experimental results show that when the label noise ratio is below 40% , both models experience only slight and negligible decreases in accuracy. When the noise ratio increases to 50% , the accuracy of the complete model falls sharply to approximately 75% due to the vulnerability of intra-flow byte features to label corruption, whereas IF^2-STG-Inter model, which relies solely on inter-flow structural features, maintains around 95% accuracy. This result further confirms that the decision-making mechanism of IF^2-STG-Inter relies primarily on robust spatio-temporal structural features rather than a strong assumption of label homogeneity, thereby demonstrating excellent adaptability to labeling noise.

In summary, although our method is proposed based on the homogeneous graph assumption, its core capability—modeling the spatio-temporal relationships of network flows—remains highly effective in real-world scenarios involving mixed traffic and labeling noise. This strengthens the feasibility of deploying our approach in practical network environments.

VI. CONCLUSION

This paper presents IF^2-STG, a novel detection framework that overcomes the limitations of existing IDS by integrating

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intra-flow features with inter-flow spatio-temporal relationships. The fundamental contribution lies in introducing an interpretable ISTG, a novel structured representation for extracting inter-flow features. By defining a two-dimensional edge vector based on time intervals and IP-set associations, the ISTG transforms implicit inter-flow collaborative patterns into explicit and analyzable structures. The model processes intra-flow features and ISTG-based inter-flow features in parallel, subsequently fusing them into a unified representation. Evaluations on three benchmark datasets show that the proposed IF^2-STG achieves over 99% classification accuracy.

More importantly, the ISTG-based model exhibits several breakthrough advantages: first, the ISTG itself is directly visualizable, providing inherent interpretability for understanding the spatio-temporal collaboration patterns of attacks such as DDoS; second, explicit structural modeling of spatio-temporal relationships endows the learned features with stronger generalization capability, leading to outstanding cross-dataset performance; finally, the model maintains good robustness against mixed traffic and label noise. Thus, this work not only provides an effective intrusion detection method but also offers a novel tool for analyzing interactive behaviors between flows.

However, the proposed method is primarily suitable for offline detection. In real-time environments, it remains challenging to instantly acquire complete flows for accurate feature extraction and graph construction. To enable online detection, future work will focus on three key challenges: First, designing a time-window-based flow sampling mechanism to process continuous packet streams. Second, addressing the explosion of graph scale in high-speed networks, we plan to partition flows within each window into manageable subgraphs with a controlled node count. Third, handling incomplete flows within single windows by introducing a window memory module, inspired by PPT-GNN [65], to correlate the same flow across consecutive windows, thereby constructing more accurate continuous spatio-temporal graphs.

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